ALARM: Alert Language-Aware Reasoning Model

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Abstract

Information processing has become a challenging task due to the unending quantity of news that is published each day online. We introduce ALARM, a framework of automated news alert generation that combines classical NLP algorithms with state-of-the-art technologies. Its hybrid architecture, combining NER, SA, image captioning, and LLM alert generation, enables a higher performance at creating relevant alerts and reduce unwanted hallucinations. Our system has achieved a 0.8082 accuracy at creating relevant news alerts, and a 3.98/5 score at "article helpfulness" in our Expert-Based Qualitative Assessment.

1 Introduction

The current information ecosystem is saturated with a continuous stream of breaking news, making it increasingly difficult for analysts and decision-makers to remain aware of the events that truly matter. This challenge has become even more relevant with the rise of prediction markets such as Polymarket and Kalshi, where participants must rapidly interpret ongoing geopolitical, economic, and social developments in order to make informed decisions.

In this context, AI-based tools have become essential for processing large volumes of information efficiently. Although Large Language Models offer strong general-purpose reasoning capabilities, we argue that they are more effective when combined with specialized components tailored to specific subtasks. Motivated by this need, we introduce ALARM, a framework for automated alert generation that integrates both classical NLP models and state-of-the-art language models to identify and structure actionable signals from news data.

More specifically, our work contributes: (i) a modular pipeline that combines image captioning, named entity recognition, sentiment analysis, and LLM-based reasoning for alert generation; (ii) a curated fine-tuning dataset for structured alert prediction; (iii) a thorough ablation study to evaluate the efectiveness of each component.

2 Related Work

News and alert generation has been an important topic in the media landscape for decades. The ability to simultaneously parse thousands of data sources has proven to be necessary to stay on top of current events.

Some early efforts in the field include NewsFinder (Dong et al. [2011]), created by researchers to find news articles about artificial intelligence in RSS fields, which relied on an SVM classifier and

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hardcoded rules to classify the topic of the article. Similarly, the Reuters Tracer algorithm (Liu et al. [2017]) was able to create news based on online data such as Tweets or other social media posts. It is based on classical machine learning methods, such as SVMs, regression and clustering. The system proved great performance, achieving over a 95% precision at detecting true news.

Deep learning techniques allowed researchers to add more advanced components to their pipelines and achieve surprising results. For instance, the study by Sufi and Alsulami [2021] integrated Entity Detection, Sentiment Analysis, and Anomaly Detection using architectures such as Convolutional Neural Networks (CNN).

With the recent rise of LLMs, many researchers have experimented with them to create useful solutions in the field. Some have created algorithms to process Spanish news (Pan et al. [2025]), while others have created specialized chatbots that are able to deliver personalized news reports (Sufi [2025]). Despite these advances, there is still a gap in hybrid frameworks that effectively combine specialized NLP components (like NER and Sentiment Analysis) with the reasoning power of LLMs to provide actionable signals for high-stakes environments like prediction markets.

3 Model

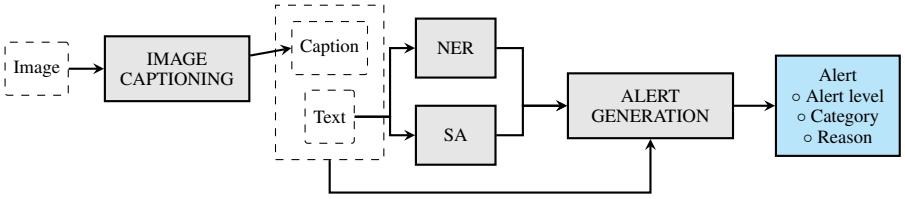
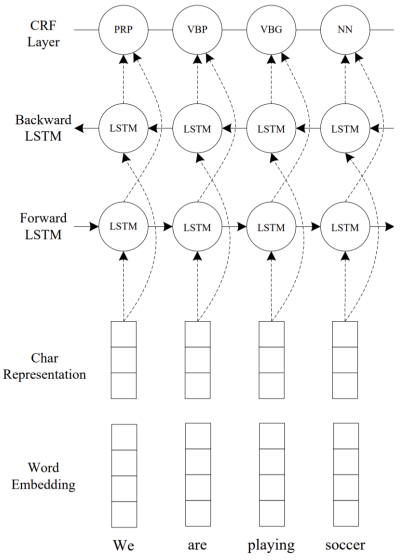


Figure 1: High-level overview of the ALARM pipeline.

3.1 Named Entity Recognition Model

For the Named Entity Recognition (NER) component of our architecture, we adopted the BiLSTM-CNN-CRF model originally proposed by Ma and Hovy [2016], both for its strong performance on the CoNLL-2003 Dataset and for its mathematical formulation, which we found interesting and solid enough to warrant direct experimentation. Our implementation is based on the reproducibility study of Ganesh and Reddy [2025], although we introduced substantial modifications to the code to correct errors and update deprecated dependencies.
The model can be seen as a pipeline composed of three successive components. First, a Convolutional Neural Network (CNN) extracts character-level morphological features for each word. These character-level vectors are concatenated with pre-trained word embeddings (GloVe-6B) and fed into a Bidirectional Long Short-Term Memory (BiLSTM) network. Finally, the hidden states are projected through a linear layer to produce local tag scores for each token, which are decoded by a Conditional Random Field (CRF) layer that models the joint probability of the label sequence, ensuring globally coherent predictions.



$$\text{Score}(x, y) = \sum_{i=1}^{n} s_i[y_i] + \sum_{i=1}^{n-1} T[y_i, y_{i+1}] + b[y_1] + e[y_n]; \quad P(y \mid x) = \frac{\exp(\text{Score}(x, y))}{\sum_{y'} \exp(\text{Score}(x, y'))}$$

During inference, the Viterbi algorithm determines the optimal tag sequence. Model training was conducted using Stochastic Gradient Descent (SGD) with momentum, employing gradient clipping

to ensure stability and a patience-based early stopping strategy to prevent overfitting. The objective function used was the negative log-likelihood.

To further optimize results, several data preprocessing techniques were implemented: the standard BIO tagging scheme was converted to BIOES for more granular label boundary detection; all numerical digits were replaced with a uniform '0' to reduce vocabulary sparsity; and all text was lowercased. Finally, sequences were padded to the maximum length. Collectively, these refinements led to a significant improvement in overall model performance.

3.2 Sentiment Analysis

The Sentiment Analysis model implemented in this project is based on a ConvBiLSTM model originally proposed by Tam et al. [2021]. This model presents a very notable architecture, since it combines the local understanding provided by CNN with the robust sequence processing of bidirectional LSTMs.

The architecture is made out of the following layers:

- **Embedding Layer**: this layer converts words into high-dimensional vectors that can be processed by neural networks. We used the pretrained Google News word2vec model, given that our goal is to process news.
- **Convolutional Layer**: this layer allows the model to extract local features by using filters. The output of the layer is sent to a Max Pooling layer, which in turn selects the most important features and reduces the dimension of the output. In other words, the model can capture the most important words in the processed text.
- **BiLSTM**: the output from the CNN structure is processed by the BiLSTM, which introduces sequence processing. This allows the model to better understand the context around the features received from the CNN. Since it is bidirectional, it can capture both preceding and succeeding context, which is an area in which a pure CNN model performs worse, according to the authors.
- **Classification layer**: the mean of the hidden states generated by the BiLSTM layer is calculated (capturing the global meaning), and it is processed through a linear layer to obtain a final prediction of the sentiment.
- **Regularisation**: the model also includes some features such as batch normalisation and dropout to avoid overfitting.

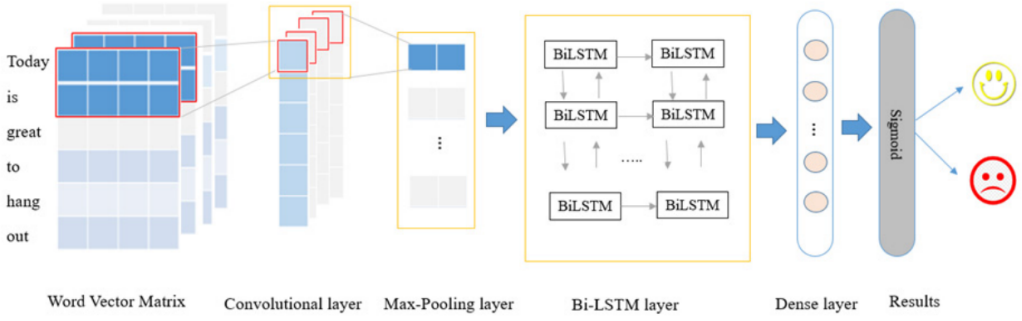


Figure 2: ConvBiLSTM architecture

Table 1 summarizes the evaluation results for both the NER and Sentiment Analysis architectures. We adopt the F_1-Score as our primary performance metric because it provides a robust, balanced measure of precision and recall. This is particularly critical for these tasks, as simple accuracy can be misleading in the presence of class imbalance, such as the prevalence of "Outside" (O) tags in NER (83%) or skewed sentiment distributions.

Table 1: F1-score comparison of the models.

Model name	F1-Score
NER Model	87.23
SA Model	83.78

3.3 Image Captioning

For the model to understand a complete news article, it is necessary for it to be able to include the information from images in articles.

For this image captioning task we chose to use the BLIP model, originally introduced in Li et al. [2022]. Hugging Face provides pretrained models related to BLIP. One of them is BLIPForConditionalGeneration[2] This model allows us to easily input an image and receive the caption.

The BLIP model is a Vision-Language Pre-training (VLP) framework that uses noisy web data by bootstrapping captions, iteratively refining pseudo-labels to improve the quality of image-text pairs used during training.

3.4 Alert-Generation Model

This section details the architecture of the proposed alert generation framework. Our approach uses a Large Language Model (LLM) designed to receive the raw article text, image caption, and the outputs of Named Entity Recognition (NER) and Sentiment Analysis (SA) models. The model employs structured output enforcement to generate JSON objects containing the alert status, severity level, alert category, and a reason to generate the alert.

We evaluated model performance by benchmarking the pre-trained Qwen2.5-7B-Instruct foundation model against a domain-specific, fine-tuned variant of it.

We use Qwen2.5-7B-Instruct as the base model due to its strong instruction-following capabilities. The model is further adapted using parameter-efficient fine-tuning (LoRA), implemented via QLoRA to enable memory-efficient training while preserving general language understanding. More specifically, we adapted Qwen2.5-7B-Instruct by training only 40.37M parameters out of 4.39B (0.92%), which provided a memory-efficient setup well suited to the size of our dataset and helped reduce the risk of overfitting. Training used 1,164 prompts for optimization and 145 for validation. We also applied label masking so that the loss was computed only on the assistant JSON completion, rather than on the instruction and input tokens, forcing the model to learn the structured output task itself. The resulting supervised-token ratio was 1.4% on the training split and 1.3% on the validation split, which is consistent with our setting of long contextual inputs and short JSON targets. Finally, inference was constrained to return a valid balanced JSON object, improving consistency and the reliability of downstream evaluation.

4 Data

4.1 Named Entity Recognition Dataset

The CoNLL [2003] dataset was selected to train our Named Entity Recognition (NER) model. We chose this corpus because its widespread adoption ensures that our results are comparable with established state-of-the-art baselines. The model was trained using the English sub-corpus, which contains 22,137 annotated sequences.

4.2 Sentiment Analysis Dataset

The dataset used to train our ConvBiLSTM Sentiment Analysis model was Sentiment140 (Go et al. [2009]). We chose this dataset as it has 1.6M examples and it is well established in Twitter Sentiment Analysis tasks.

[2]See the Hugging Face documentation for `BLIPForConditionalGeneration`.

Although the information on Kaggle [3] indicates this dataset has 3 labels: 0 for negative, 2 for neutral and 4 for positive, we only found labels 0 and 4 in the dataset.

4.3 Fine-tuning Dataset for the Alert Generation Model

To facilitate the fine-tuning of the alert generation model, we created a specialized dataset leveraging the VisualNews [2021] corpus. We extracted the article text and image captions, processing them through the NER and SA models to append the resulting outputs as supplementary features. From the initial collection of 100,000 entries, a subset of 1,500 rows was selected for labeling.

For these records, we employed Gemini-3.1-Flash-Lite to synthesize ground-truth alerts, adhering to the pre-defined JSON schema used by the final model. An analysis was conducted on the resulting dataset to ensure it was balanced and clean, providing a robust foundation for the final model.

The dataset consists of 1,455 news articles, of which 696 (47.8%) contain alerts and 759 (52.2%) do not. Each alert includes not only a category label but also a brief explanation clarifying why the alert was generated. Alerts are further categorized into multiple risk types, with the most frequent being reputation risk and geopolitical events.

5 Experiments

This section outlines the experimental framework employed to evaluate the performance of the alert model. We use a combination of quantitative and qualitative analysis to ensure a robust assessment of the model's capabilities.

5.1 Expert-Based Qualitative Assessment

Current state-of-the-art methodologies for evaluating Large Language Models (LLMs) and other Deep Learning architectures emphasize the necessity of human-centric evaluation (Chaudhari et al. [2018], Tam et al. [2024]). To this end, we established an evaluation pipeline in which four domain experts (represented by us) performed a review of the model's outputs.

The evaluation framework is based on three metrics, each quantified using a 5 point Likert scale:

Table 2: Qualitative evaluation rubric used for expert assessment.

Metric	**Description**	**1 (Poor)**	**5 (Excellent)**
Factuality	Is the information truthful and free of hallucinations?	Contains serious factual errors.	100% truthful and verifiable.
Instruction Following	Does it follow format, tone, and constraints?	Ignored the instructions.	Followed every requirement.
Helpfulness	How much value does it provide to the end user?	Useless or needs full rework.	Really relevant alert.

The results of the expert assessment (Table 3) indicate high performance across the metrics. Both Factuality and Instruction Following yielded near-perfect scores. In contrast, the Helpfulness metric exhibited a lower mean score and high variability. This divergence is attributed to the subjective nature of the "Helpfulness" criterion in the context of news monitoring. The model often generates alerts for news that might be alarming for a general audience; however, the expert evaluators tended to rate these same alerts as having lower utility. This discrepancy highlights a difference in risk perception thresholds between the model and the experts' judgment.

[3] `https://www.kaggle.com/datasets/kazanova/sentiment140`

Table 3: Qualitative evaluation scores.

	mean	**min**	**max**
factuality_mean	4.82 ± 0.37	4.0	5.0
instruction_following_mean	4.86 ± 0.56	2.5	5.0
helpfulness_mean	3.98 ± 1.48	1.0	5.0

We quantified the level of consensus among the four experts using the Intraclass Correlation Coefficient (ICC). We opted to use the ICC to account for the ordinal nature of our evaluation scales. The analysis, conducted across 20 samples, yielded consensus coefficients of 0.889 in the Factuality metric, 0.912 in Instruction Following and 0.866 in Helpfulness. These values indicate excellent agreement among the evaluators. This high level of statistical consensus confirms the robustness of our evaluation framework and ensures that the qualitative metrics were applied consistently across the study.

5.2 Ablation Study

To quantify the individual contribution of each module to the overall system performance, we conducted an ablation study. This evaluation involved a series of controlled experiments where the NER, SA and Image Caption components were sequentially removed from the pipeline.

Table 4 summarizes this section's findings. As anticipated, the removal of most modules led to performance degradation, in particular the image captioning and sentiment analysis modules, suggesting that they contribute meaningfully to alert generation. However, excluding the NER component did not have a significant effect on the performance, likely due to two primary factors:

- **Context Window Saturation:** The high verbosity of the NER output significantly increases the input length. For smaller models such as Qwen2.5-7B, this excessive context likely introduced noise and worsened processing efficiency.
- **Redundancy:** Modern LLMs possess inherent entity recognition capabilities. Consequently, an external NER module proved redundant, adding complexity and overhead without providing additional utility.

Table 4: Ablation results for the fine-tuned model under different module removal settings.

Model	**Acc.**	**Alert**					**No Alert**		
		Prec.	**Rec.**	**F1**	**Level Acc.**	**Cat. Acc.**	**Prec.**	**Rec.**	**F1**
Fine-tuned (full)	**0.8082**	0.907	**0.681**	**0.778**	—	—	**0.750**	0.932	**0.831**
Fine-tuned (w/o caption)	0.7877	0.902	0.639	0.748	0.3333	0.5972	0.726	0.932	0.817
Fine-tuned (w/o NER)	**0.8014**	**0.939**	0.639	**0.760**	**0.3472**	**0.6528**	**0.732**	**0.959**	**0.830**
Fine-tuned (w/o SA)	0.7055	**0.939**	0.431	0.590	0.1806	0.5694	0.637	**0.973**	0.770

5.3 Determinism Analysis

To evaluate the consistency of the model's outputs, we conducted an experiment processing $N = 146$ articles through the complete pipeline across two independent iterations. A comparison of the runs yielded an inter-run accuracy of 1.0 for alert generation, demonstrating that the model is perfectly consistent in identifying relevant articles. For categorical classification, we observed an accuracy of 0.85, indicating minor variations in label assignment between runs. Despite these slight fluctuations in categorization, the high level of agreement in alert generation confirms the architectural robustness of the model and ensures the reproducibility of our experimental results.

6 Results

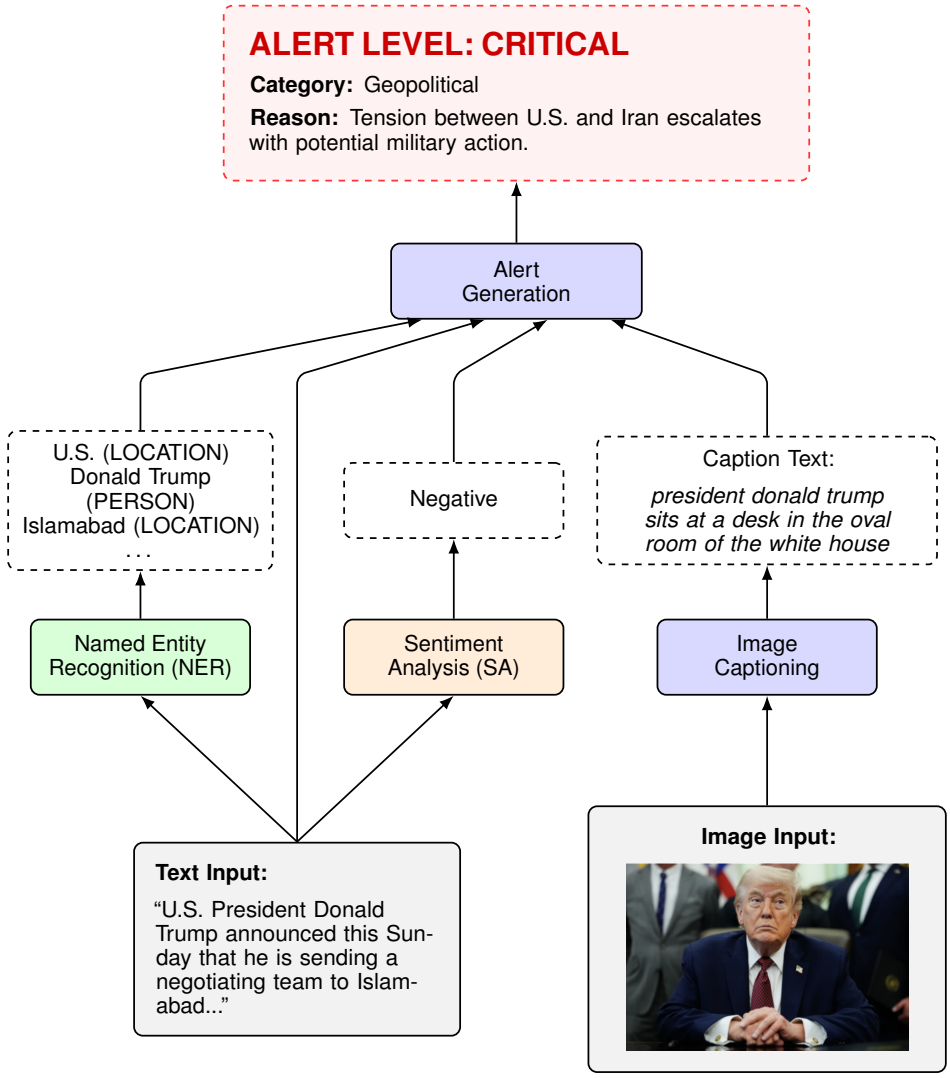
6.1 Alert Generation Model

Table 5 compares the base and fine-tuned versions of the alert generation model on the alert classification task. The fine-tuned model achieves higher overall accuracy, while showing a different precision-recall trade-off across the alert and no-alert classes.

Table 5: Comparison between the base and fine-tuned models on the alert classification task.

Model	Acc.	Lat. (s)	Alert			No Alert		
			Prec.	Rec.	F1	Prec.	Rec.	F1
Base (Qwen2.5-7B)	0.7945	**1.762**	0.776	**0.819**	**0.797**	**0.814**	0.770	0.792
Fine-tuned (ours)	**0.8082**	2.156	**0.907**	0.681	0.778	0.750	**0.932**	**0.831**

Here is an example of an output of the ALARM model:



7 Conclusion

In this paper, we have thoroughly explored the architecture of our ALARM framework.

Our Sentiment Analysis obtained an 83.78% F1-Score in the selected dataset's test split, which demonstrates a solid performance at the task. In addition to this, our ablation study proved that this module is key for the entire architecture.

The Named Entity Recognition model was able to reach an 87.23% F1-Score. Nevertheless, our experiments showed that, because of context window saturation and redundancy, the complete pipeline performed slightly better without this component.

As for Image Captioning, the BLIP model provides reliable image captions. Even though most article texts are sufficiently descriptive of the news, we did observe a slight improvement in accuracy with this component.

Finally, we fine-tuned an LLM based on Qwen2.5. In our experiments, multiple training iterations led to a modest but consistent improvement in performance, suggesting that the proposed approach is effective while still leaving significant room for further development.

We believe that training on a larger and more diverse dataset would improve the model's robustness and generalization capabilities. As future work, we plan to continue refining the system through additional fine-tuning and expanded data collection.

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